

Least-squares FIR design:

- Given desired signed-magnitude response $D(\omega)$, we will now design the length- L GLP filter $h[n]$ that minimizes the squared response error

error energy of
our approximation

$$\mathcal{E}_2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \underbrace{|D(e^{j\omega})|}_{\text{desired}} - \underbrace{H(e^{j\omega})}_{\text{implementation}}|^2 d\omega.$$

- For a length- L GLP filter, the desired freq response is

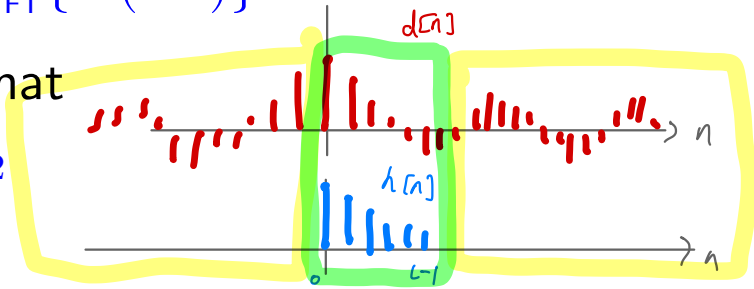
$$D(e^{j\omega}) = D(\omega)e^{-j(\omega \frac{L-1}{2} + \phi_0)},$$

from which we can obtain the desired impulse response

$$d[n] = \mathcal{F}_{\text{DTFT}}^{-1}\{D(e^{j\omega})\}.$$

- Parseval's theorem implies that

$$\begin{aligned} \mathcal{E}_2 &= \sum_{n=-\infty}^{\infty} |d[n] - h[n]|^2 \\ &= \sum_{n=-\infty}^{-1} |d[n]|^2 + \sum_{n=0}^{L-1} |d[n] - h[n]|^2 + \sum_{n=L}^{\infty} |d[n]|^2 \end{aligned}$$



which is minimized by

$$h[n] = d[n] \text{ for } n = 0, \dots, L-1.$$

$\Rightarrow$ rectangular-window method is least-squares optimal!

- But rectangular-window designs have very large ripples!
Why are they least-squares optimal? The LS criterion doesn't include

don't-care regions, and so the filter worries most about transition-band error.

Weighted least-squares FIR design:

- The weighted-LS criterion allows more control over ripple performance:

$$\mathcal{E}_W = \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) |D(e^{j\omega}) - H(e^{j\omega})|^2 d\omega.$$

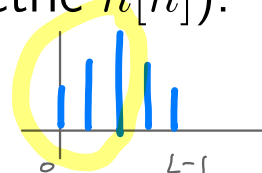
assign zero weight on transition band.

- For length- L GLP, $D(e^{j\omega}) = D(\omega)e^{-j(\omega\frac{L-1}{2} + \phi_0)}$, so

$$\begin{aligned} \mathcal{E}_W &= \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) |D(\omega)e^{-j(\omega\frac{L-1}{2} + \phi_0)} - H(e^{j\omega})|^2 d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) |D(\omega) - H(e^{j\omega})e^{j(\omega\frac{L-1}{2} + \phi_0)}|^2 d\omega. \end{aligned}$$

- For Type-I GLP filters (odd L , $\phi_0 = 0$, symmetric $h[n]$):

$$\begin{aligned} H(e^{j\omega})e^{j(\omega\frac{L-1}{2} + \phi_0)} &= \sum_{n=0}^{L-1} h[n]e^{-j\omega(n - \frac{L-1}{2})} \\ &= (h[0] + h[L-1])\cos(\omega\frac{L-1}{2}) + j(h[0] - h[L-1])\sin(\omega\frac{L-1}{2}) \\ &\quad + (h[1] + h[L-2])\cos(\omega\frac{L-3}{2}) + j(h[1] - h[L-2])\sin(\omega\frac{L-3}{2}) \\ &\quad + \dots + h[(L-1)/2] \\ &= h[(L-1)/2] + \sum_{n=0}^{\frac{L-1}{2}-1} 2h[n]\cos(\omega(\frac{L-1}{2} - n)) \quad \text{for real-valued } h[n] \\ &= \sum_{n=0}^{\frac{L-1}{2}} c_n \cos(\omega(\frac{L-1}{2} - n)) \quad \text{with } c_n = \begin{cases} 2h[n] & \text{for } n = 0 \dots \frac{L-1}{2} - 1 \\ h[n] & \text{for } n = \frac{L-1}{2} \end{cases} \end{aligned}$$



*parametric model of
freq response*

Weighted least-squares FIR design (cont.):

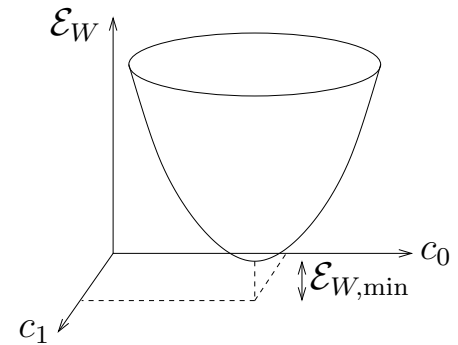
For real length- L Type-I GLP filters, the WLS criterion is:

$$\mathcal{E}_W = \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) \left[D(\omega) - \sum_{n=0}^{(L-1)/2} c_n \cos\left(\omega\left(\frac{L-1}{2} - n\right)\right) \right]^2 d\omega$$

function of $\begin{bmatrix} c_0 \\ \vdots \\ c_{\frac{L-1}{2}} \end{bmatrix}$

which is a quadratic function of the coefficients $\{c_n\}_{n=0}^{(L-1)/2}$.

At the optimal solution, the derivatives $\left\{ \frac{\partial \mathcal{E}_W}{\partial c_l} \right\}_{l=0}^{(L-1)/2}$ must equal zero.



These derivatives are:

$$\begin{aligned} \frac{\partial \mathcal{E}_W}{\partial c_l} &= \frac{1}{2\pi} \int_{-\pi}^{\pi} W(\omega) \frac{\partial}{\partial c_l} \left[D(\omega) - \sum_{n=0}^{(L-1)/2} c_n \cos\left(\omega\left(\frac{L-1}{2} - n\right)\right) \right]^2 d\omega \\ &= \frac{-2}{2\pi} \int_{-\pi}^{\pi} W(\omega) \cos\left(\omega\left(\frac{L-1}{2} - l\right)\right) \left[D(\omega) - \sum_{n=0}^{(L-1)/2} c_n \cos\left(\omega\left(\frac{L-1}{2} - n\right)\right) \right] d\omega \\ &= -\frac{1}{\pi} \int_{-\pi}^{\pi} W(\omega) \cos\left(\omega\left(\frac{L-1}{2} - l\right)\right) D(\omega) d\omega \\ &\quad + \sum_{n=0}^{(L-1)/2} c_n \frac{1}{\pi} \int_{-\pi}^{\pi} W(\omega) \cos\left(\omega\left(\frac{L-1}{2} - l\right)\right) \cos\left(\omega\left(\frac{L-1}{2} - n\right)\right) d\omega. \end{aligned}$$

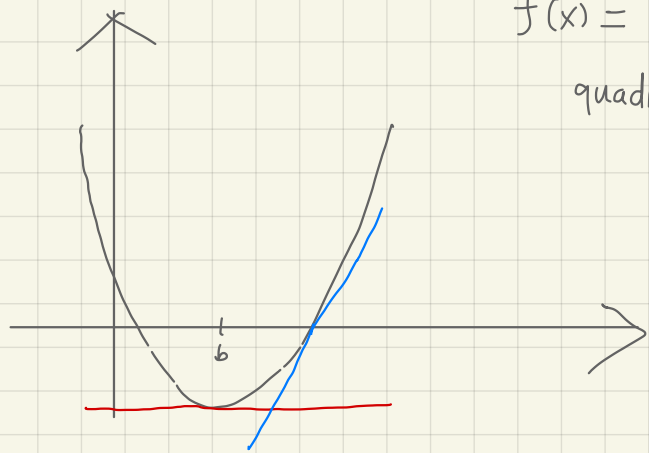
Zeroing gives $\frac{L-1}{2} + 1$ linear equations in $\frac{L-1}{2} + 1$ unknowns:

$$0 = -p_l + \sum_{n=0}^{(L-1)/2} c_n Q_{l,n} \text{ for } l = 0 \dots \frac{L-1}{2} \Leftrightarrow \mathbf{p} = \mathbf{Q}\mathbf{c}$$

which can be solved via $\mathbf{c} = \mathbf{Q}^{-1}\mathbf{p}$.

$$f(x) = a(x-b)^2 + c \quad a > 0$$

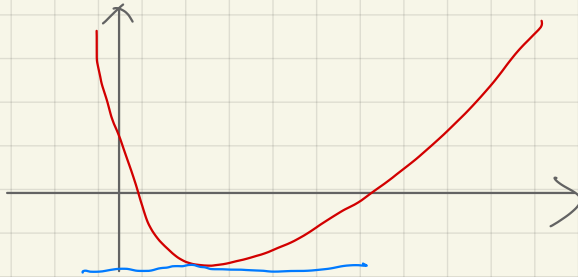
quadratic function



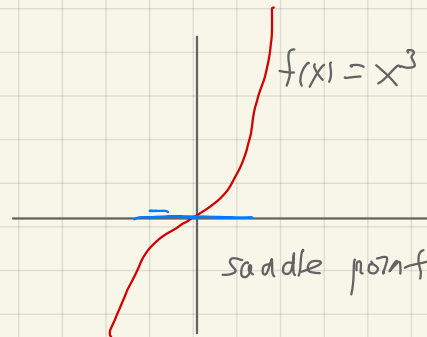
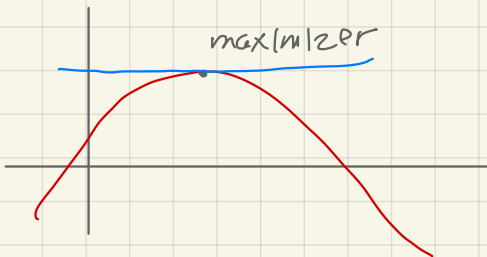
$$f'(x) = \lim_{\Delta \rightarrow 0} \frac{f(x+\Delta) - f(x)}{\Delta}$$

$$f'(x) = 2a(x-b)$$

$$f'(b) = 0 \quad f''(x) = 2a$$



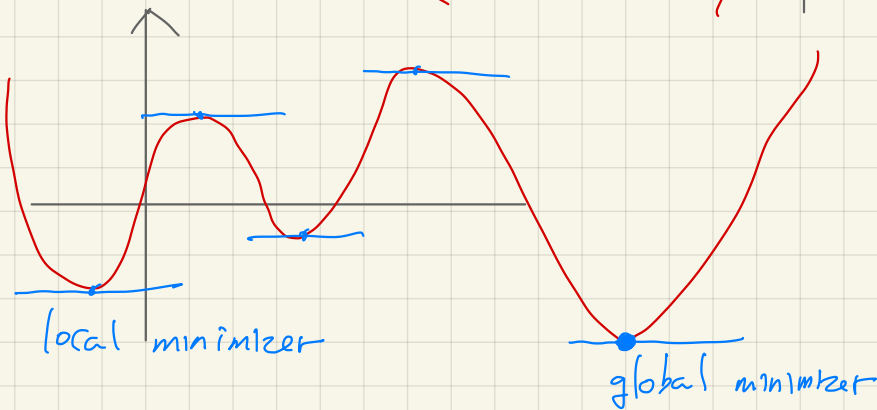
$$f'(x) = 0 ?$$



$$f(x) = x^3$$

$$f'(x) = 3x^2$$

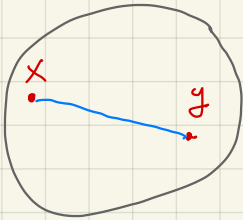
$$f'(0) = 0$$



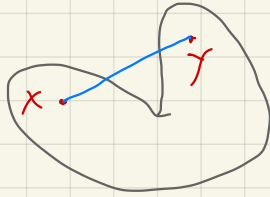
Convexity

S is a convex set if $\lambda x + (1-\lambda)y \in S$ for all $0 \leq \lambda \leq 1$ all $x, y \in S$
convex combination.

$S \in \mathbb{R}^2$



convex

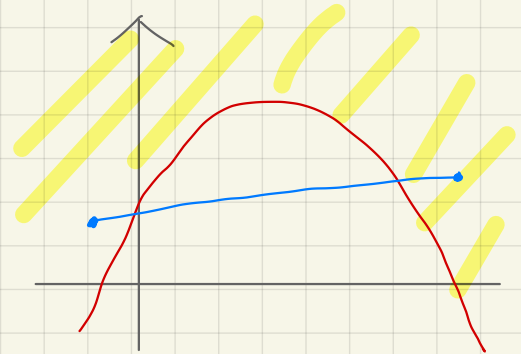
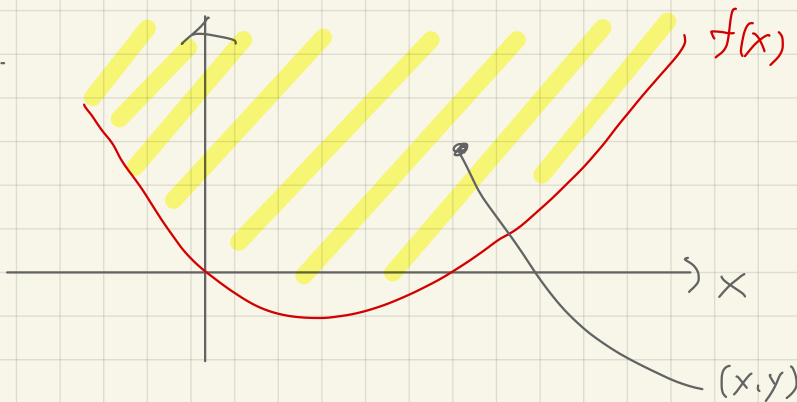


not convex

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$\text{epi}(f) = \{ (x, y) : y \geq f(x) \}$$

epigraph.



f is a convex function iff $\text{epi}(f)$ is a convex set.

If f is a convex function, and differentiable, then
 x_0 is a minimizer if $f'(x_0) = 0$

If f is twice differentiable,

f is convex if $f''(x) \geq 0$ for all x .